

## LAMBERTS BAY NORTIER, South Africa

WMO: 686130

Lat: 32.035S

Lon: 18.332E

Elev: 307

StdP: 14.53

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 42.2       | 43.9 | 26.5                          | 20.7 | 68.5 | 29.9 | 24.3 | 67.6 | 20.0                  | 63.9 | 17.3 | 62.7 | 6.1                   | 100  | 0.419 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 15.5                   | 90.4            | 64.8 | 85.8 | 63.3 | 82.0 | 62.2 | 68.8                | 80.2 | 67.4 | 77.3 | 66.3 | 75.0 | 11.6                 | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.7                            | 95.8 | 71.0 | 64.3 | 91.1 | 69.9 | 63.1 | 87.5 | 69.2 | 33.2          | 80.5 | 32.0 | 77.5 | 31.1 | 75.2 | 75.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 19.4          | 16.7 | 14.7 | DB                         | 39.2 | 99.4               | 1.8 | 3.1                                                | 37.9 | 101.6      | 36.9 | 103.4      | 35.9 | 105.1      | 34.6 | 107.3 | (4) |
| (5)               |      |      | WB                         | 37.9 | 71.6               | 1.7 | 1.1                                                | 36.6 | 72.3       | 35.6 | 73.0       | 34.7 | 73.6       | 33.4 | 74.4  | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 61.9   | 67.3  | 67.9  | 66.2  | 64.5  | 60.2  | 57.1  | 56.3  | 55.6  | 58.5  | 61.4  | 63.3  | 65.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 6.81   | 3.62  | 4.49  | 4.78  | 7.27  | 6.36  | 5.54  | 6.62  | 5.07  | 5.64  | 5.61  | 4.24  | 3.72 | (7)  |
| (8)  |                                                                                  | HDD50       | 11     | 0     | 0     | 0     | 0     | 0     | 2     | 7     | 2     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD65       | 1643   | 12    | 10    | 32    | 95    | 185   | 250   | 287   | 299   | 215   | 144   | 80    | 34   | (9)  |
| (10) |                                                                                  | CDD50       | 4370   | 536   | 501   | 503   | 435   | 317   | 216   | 203   | 177   | 255   | 352   | 398   | 477  | (10) |
| (11) |                                                                                  | CDD65       | 527    | 82    | 91    | 70    | 80    | 37    | 14    | 18    | 9     | 20    | 31    | 29    | 46   | (11) |
| (12) |                                                                                  | CDH74       | 3629   | 388   | 567   | 497   | 660   | 267   | 75    | 88    | 64    | 194   | 312   | 242   | 275  | (12) |
| (13) |                                                                                  | CDH80       | 1227   | 108   | 210   | 169   | 249   | 83    | 14    | 11    | 12    | 79    | 121   | 77    | 94   | (13) |
| (14) | Wind                                                                             | WSAvg       | 7.1    | 7.3   | 7.0   | 6.8   | 6.8   | 6.4   | 7.1   | 7.1   | 6.9   | 7.1   | 7.6   | 7.9   | 7.4  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 8.6    | 0.2   | 0.1   | 0.2   | 0.6   | 1.1   | 1.4   | 1.6   | 1.3   | 0.7   | 0.4   | 0.4   | 0.3  | (15) |
| (16) |                                                                                  | PrecMax     | 13.6   | 0.9   | 0.6   | 0.7   | 1.4   | 3.5   | 2.8   | 5.2   | 2.8   | 2.1   | 1.6   | 1.9   | 1.5  | (16) |
| (17) |                                                                                  | PrecMin     | 4.6    | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.3   | 0.4   | 0.0   | 0.0   | 0.0   | 0.0  | (17) |
| (18) |                                                                                  | PrecStd     | 2.4    | 0.3   | 0.2   | 0.2   | 0.4   | 0.9   | 0.7   | 1.3   | 0.7   | 0.5   | 0.4   | 0.4   | 0.4  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 88.8  | 95.4  | 93.0  | 94.0  | 88.3  | 82.4  | 81.0  | 81.4  | 88.8  | 92.4  | 87.2  | 91.1 | (19) |
| (20) |                                                                                  |             | MCWB   | 66.1  | 67.0  | 65.7  | 65.5  | 61.3  | 58.8  | 58.5  | 57.4  | 60.5  | 63.6  | 62.1  | 65.5 | (20) |
| (21) |                                                                                  | 2%          | DB     | 82.6  | 87.3  | 84.7  | 86.8  | 81.2  | 75.5  | 76.0  | 73.8  | 78.9  | 82.8  | 81.0  | 80.9 | (21) |
| (22) |                                                                                  |             | MCWB   | 64.9  | 66.1  | 64.9  | 62.3  | 58.8  | 56.3  | 56.0  | 55.7  | 56.6  | 60.9  | 61.3  | 63.3 | (22) |
| (23) |                                                                                  | 5%          | DB     | 77.6  | 81.2  | 79.6  | 81.6  | 75.7  | 70.4  | 71.5  | 68.5  | 71.9  | 76.3  | 75.1  | 75.8 | (23) |
| (24) |                                                                                  |             | MCWB   | 64.5  | 64.8  | 63.6  | 61.3  | 57.1  | 54.6  | 54.3  | 53.2  | 55.8  | 58.7  | 60.3  | 63.5 | (24) |
| (25) |                                                                                  | 10%         | DB     | 74.7  | 76.2  | 75.0  | 76.5  | 70.2  | 65.9  | 66.9  | 63.4  | 66.8  | 70.9  | 71.3  | 73.0 | (25) |
| (26) |                                                                                  |             | MCWB   | 64.4  | 64.4  | 63.6  | 59.8  | 56.9  | 53.6  | 53.1  | 53.6  | 55.7  | 58.2  | 60.7  | 63.1 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 69.9  | 71.1  | 70.3  | 67.9  | 63.3  | 60.9  | 60.2  | 60.4  | 62.0  | 66.4  | 67.2  | 68.5 | (27) |
| (28) |                                                                                  |             | MCD B  | 81.7  | 85.1  | 82.3  | 89.2  | 79.0  | 72.2  | 73.4  | 71.7  | 77.9  | 84.4  | 76.0  | 79.4 | (28) |
| (29) |                                                                                  | 2%          | WB     | 67.9  | 68.9  | 68.0  | 65.0  | 61.2  | 58.6  | 57.9  | 58.1  | 59.5  | 62.8  | 64.7  | 66.6 | (29) |
| (30) |                                                                                  |             | MCD B  | 75.8  | 80.6  | 77.9  | 81.2  | 72.4  | 66.7  | 68.1  | 67.0  | 70.1  | 73.9  | 72.3  | 75.6 | (30) |
| (31) |                                                                                  | 5%          | WB     | 66.6  | 67.1  | 66.3  | 63.0  | 60.0  | 57.4  | 56.5  | 56.4  | 58.3  | 61.2  | 63.1  | 65.2 | (31) |
| (32) |                                                                                  |             | MCD B  | 74.1  | 76.7  | 74.8  | 75.3  | 69.0  | 63.8  | 64.8  | 63.5  | 66.7  | 70.0  | 70.5  | 72.8 | (32) |
| (33) |                                                                                  | 10%         | WB     | 65.5  | 65.7  | 64.9  | 61.6  | 58.6  | 56.4  | 55.1  | 55.2  | 57.2  | 60.0  | 62.0  | 64.1 | (33) |
| (34) |                                                                                  |             | MCD B  | 72.5  | 73.7  | 72.4  | 71.6  | 65.7  | 62.0  | 62.4  | 60.6  | 63.9  | 67.7  | 69.1  | 70.6 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.9  | 15.5  | 15.1  | 16.0  | 14.5  | 14.2  | 15.7  | 15.1  | 15.9  | 16.2  | 15.2  | 15.1 | (35) |
| (36) |                                                                                  |             | MCDBR  | 19.9  | 22.8  | 22.2  | 24.0  | 25.0  | 22.0  | 22.7  | 24.3  | 24.5  | 24.7  | 22.4  | 19.4 | (36) |
| (37) |                                                                                  |             | MCWBR  | 8.2   | 8.4   | 8.4   | 8.9   | 9.3   | 9.2   | 9.7   | 10.6  | 9.8   | 10.4  | 9.4   | 8.4  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 14.9  | 16.4  | 15.9  | 18.6  | 17.8  | 15.2  | 17.3  | 16.7  | 17.7  | 17.3  | 15.5  | 15.6 | (38) |
| (39) |                                                                                  |             | MCWBR  | 7.7   | 8.6   | 8.5   | 8.4   | 9.2   | 9.2   | 9.8   | 10.1  | 9.7   | 9.3   | 8.6   | 8.3  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.334  | 0.336 | 0.331 | 0.316 | 0.308 | 0.300 | 0.304 | 0.323 | 0.333 | 0.342 | 0.327 | 0.332 | (40) |      |
| (41) |                                                                                  | taud        | 2.544  | 2.546 | 2.556 | 2.596 | 2.599 | 2.587 | 2.567 | 2.511 | 2.453 | 2.448 | 2.537 | 2.539 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 319    | 312   | 303   | 291   | 278   | 272   | 276   | 284   | 297   | 306   | 319   | 320   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 35     | 34    | 31    | 27    | 24    | 23    | 24    | 29    | 34    | 36    | 35    | 35    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2638   | 2389  | 1964  | 1465  | 1057  | 904   | 1006  | 1246  | 1674  | 2148  | 2516  | 2686  | (44) |      |
| (45) |                                                                                  | RadStd      | 69     | 64    | 58    | 71    | 66    | 48    | 52    | 58    | 72    | 89    | 78    | 104   | (45) |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (5 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | -1.87 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page